Classical Mechanics from Zero

Lecture Notes for Computer Science Students with Weak Mathematical Background

Prepared as beginner-friendly university lecture notes

May 16, 2026

Goal. These notes teach the core ideas of elementary classical mechanics slowly and explicitly. The reader is not expected to be comfortable with physics. The notes explain the meaning of each symbol, show all important derivations, and include full examples.

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1 What mechanics is about

Mechanics

Mechanics is the part of physics that studies motion and the causes of motion.
In elementary mechanics we usually ask questions such as:

- Where is an object now?
- How fast is it moving?
- Is its velocity changing?
- What forces act on it?
- How much energy or momentum does it have?
- What will happen next?

For computer science students, mechanics is useful because it appears in:

- computer graphics and animation,
- game engines,
- robotics,
- physics simulations,
- optimization and control,
- numerical methods,
- machine learning for physical systems.

A simple game engine already needs mechanics. If a ball is thrown, the program must update its position. If a character jumps, gravity must be simulated. If two objects collide, their velocities should change in a plausible way.

The basic computational question

Mechanics often becomes an algorithmic problem:

$$\text{current state} + \text{rules of motion} \longrightarrow \text{next state}.$$

This is why mechanics is a natural topic for programmers.

1.1 The minimum vocabulary

The most important quantities are listed below.

Quantity	Meaning	Common unit
Time t	When something happens	second s
Position x, $\boldsymbol{r}$	Where an object is	meter m
Velocity v, $\boldsymbol{v}$	How fast and in what direction position changes	m/s
Acceleration a, $\boldsymbol{a}$	How fast velocity changes	$\mathrm{m/s^2}$
Mass m	How hard it is to change motion	kilogram kg
Force F, $\boldsymbol{F}$	Interaction that changes motion	newton N
Energy E	Capacity to do work or cause change	joule J
Momentum p, $\boldsymbol{p}$	Mass times velocity	kg m/s

Do not memorize symbols without meaning

A formula is not a magic spell. Every symbol has a meaning and a unit. If the units do not match, the formula is probably being used incorrectly.

2 Units, dimensions, and sanity checks

2.1 Units are part of the number

Writing

$$x = 5$$

for a position is incomplete. We need a unit:

$$x = 5\,\mathrm{m}.$$

A number without a unit is not enough in physics. Is it 5 meters, 5 kilometers, 5 seconds, or 5 pixels?

Dimension

The dimension of a quantity describes its physical type. For example:

- length has dimension $[L]$,
- time has dimension $[T]$,
- mass has dimension $[M]$,
- velocity has dimension $[L]/[T]$,
- acceleration has dimension $[L]/[T]^2$.

2.2 Example: checking a formula

Suppose somebody says that distance traveled under constant speed is

$$s = vt.$$

Check units:

$$[vt] = \frac{\mathrm{m}}{\mathrm{s}} \cdot \mathrm{s} = \mathrm{m}.$$

The result has units of meters, so the formula is dimensionally reasonable.

Now suppose somebody says:

$$s = v + t.$$

This is nonsense because v has units m/s, while t has units s. We cannot add quantities of different physical type.

Common beginner mistake

Never add meters to seconds, or meters per second to seconds. Addition and subtraction require the same unit type.

3 Scalars and vectors

3.1 Scalars

A scalar is a quantity described by one number and a unit.

Examples:

$$m = 2\,\mathrm{kg}, \qquad t = 5\,\mathrm{s}, \qquad T = 300\,\mathrm{K}.$$

Mass and time are scalars.

3.2 Vectors

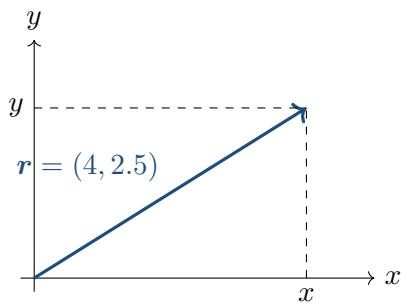
A vector has magnitude and direction. In one dimension, a vector can be represented by a signed number. In two or three dimensions, it has components.

For example, a two-dimensional position vector is

$$\boldsymbol{r} = (x, y).$$

A velocity vector is

$$\boldsymbol{v} = (v_x, v_y).$$



3.3 Length of a vector

For a vector $\boldsymbol{v} = (v_x, v_y)$, its length is

$$|\boldsymbol{v}| = \sqrt{v_x^2 + v_y^2}.$$

This is just the Pythagorean theorem.

Full example: speed from velocity components

Let

$$\boldsymbol{v} = (3, 4)\,\mathrm{m/s}.$$

The speed is the length of the velocity vector:

$$|\boldsymbol{v}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5\,\mathrm{m/s}.$$

So the object moves with speed $5\,\mathrm{m/s}$, while its velocity points in the direction $(3, 4)$.

Checkpoint 1

Compute the length of $\boldsymbol{v} = (6, 8)\,\mathrm{m/s}$. Answer: $10\,\mathrm{m/s}$.

4 Position, displacement, velocity, and speed

4.1 Position

In one dimension, position is described by a number x. For example:

$$x = 10\,\mathrm{m}.$$

This means the object is 10 meters from the chosen origin.

The origin is the point we decide to call zero. It is not necessarily special in nature; it is a coordinate choice.

4.2 Displacement

Displacement is change in position:

$$\Delta x = x_{\text{final}} - x_{\text{initial}}.$$

The symbol Δ means "change in".

Displacement

An object moves from

$$x_0 = 2\,\mathrm{m}$$

to

$$x_1 = 9\,\mathrm{m}.$$

Then

$$\Delta x = x_1 - x_0 = 9\,\mathrm{m} - 2\,\mathrm{m} = 7\,\mathrm{m}.$$

If an object moves from $x = 9\,\mathrm{m}$ to $x = 2\,\mathrm{m}$, then

$$\Delta x = 2\,\mathrm{m} - 9\,\mathrm{m} = -7\,\mathrm{m}.$$

The negative sign means motion in the negative direction.

4.3 Average velocity

Average velocity is displacement divided by elapsed time:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}.$$

Average velocity

An object moves from $x = 3\,\mathrm{m}$ at $t = 0\,\mathrm{s}$ to $x = 15\,\mathrm{m}$ at $t = 4\,\mathrm{s}$. Then

$$\Delta x = 15 - 3 = 12\,\mathrm{m},$$

$$\Delta t = 4 - 0 = 4\,\mathrm{s},$$

$$v_{\text{avg}} = \frac{12\,\mathrm{m}}{4\,\mathrm{s}} = 3\,\mathrm{m/s}.$$

4.4 Speed versus velocity

Velocity has direction. Speed does not. In one dimension:

$$\text{speed} = |v|.$$

If $v = -4\,\mathrm{m/s}$, then the speed is $4\,\mathrm{m/s}$. The minus sign tells direction, not slowness.

Common beginner mistake

Negative velocity does not mean the object is slowing down. It means it moves in the negative direction. Slowing down means the magnitude of velocity is decreasing.

5 Instantaneous velocity and the derivative idea

5.1 The problem with average velocity

Average velocity gives information about an interval of time. But we often want the velocity at one instant.

If position is a function of time,

$$x = x(t),$$

then average velocity from t to $t + \Delta t$ is

$$\frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

Instantaneous velocity is obtained by making the time interval smaller and smaller:

$$v(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

This is the derivative:

$$v(t) = \frac{\mathrm{d}x}{\mathrm{d}t}.$$

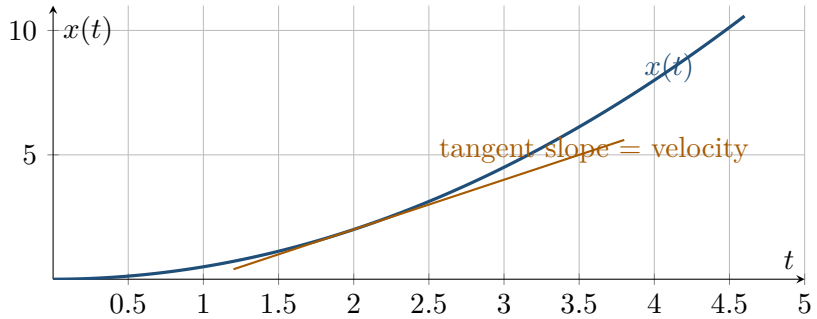
Velocity as a derivative

If $x(t)$ is position as a function of time, then velocity is

$$v(t) = x'(t) = \frac{\mathrm{d}x}{\mathrm{d}t}.$$

5.2 Graphical meaning

Velocity is the slope of the position-time graph.



5.3 Full example

Let

$$x(t) = t^2 + 2t.$$

Then

$$v(t) = x'(t) = 2t + 2.$$

At $t = 3\,\mathrm{s}$:

$$v(3) = 2 \cdot 3 + 2 = 8\,\mathrm{m/s}.$$

Units in derivatives

If x is measured in meters and t in seconds, then $\mathrm{d}x/\,\mathrm{d}t$ has units meters per second.

6 Acceleration

Acceleration is the rate of change of velocity.

Acceleration

In one dimension,

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}.$$

Acceleration is the second derivative of position.

If velocity changes by Δv during time Δt, average acceleration is

$$a_{\mathrm{avg}} = \frac{\Delta v}{\Delta t}.$$

Acceleration from velocity

Let

$$v(t) = 4t - 1.$$

Then

$$a(t) = v'(t) = 4.$$

The acceleration is constant:

$$a = 4\,\mathrm{m/s^2}.$$

Acceleration from position

Let

$$x(t) = 3t^2 - 2t + 5.$$

Velocity is

$$v(t) = x'(t) = 6t - 2.$$

Acceleration is

$$a(t) = v'(t) = 6.$$

So acceleration is $6\,\mathrm{m/s^2}$.

6.1 Acceleration is not the same as velocity

Velocity tells how position changes. Acceleration tells how velocity changes.

An object may have:

- positive velocity and positive acceleration: moving right and speeding up,
- positive velocity and negative acceleration: moving right and slowing down,
- negative velocity and positive acceleration: moving left and slowing down,
- negative velocity and negative acceleration: moving left and speeding up.

Common beginner mistake

Do not say "acceleration is speed". Acceleration is change of velocity. A car moving at constant 100 km/h has high speed but zero acceleration.

7 Constant acceleration formulas

Many beginner problems use constant acceleration. This is important because gravity near Earth is approximately constant.

Assume acceleration is constant:

$$a(t) = a.$$

Since acceleration is the derivative of velocity,

$$\frac{\mathrm{d}v}{\mathrm{d}t} = a.$$

Integrating with respect to time gives

$$v(t) = v_0 + at,$$

where v_0 is the initial velocity.

Since velocity is the derivative of position,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v_0 + at.$$

Integrating again:

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2.$$

Constant acceleration equations

For motion in one dimension with constant acceleration:

$$v(t) = v_0 + at,$$

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2.$$

Also,

$$v^2 = v_0^2 + 2a(x - x_0).$$

7.1 Deriving the formula without time

We derive

$$v^2 = v_0^2 + 2a(x - x_0).$$

Start from

$$v = v_0 + at.$$

So

$$t = \frac{v - v_0}{a}.$$

For constant acceleration, average velocity is

$$v_{\text{avg}} = \frac{v_0 + v}{2}.$$

Displacement is average velocity times time:

$$x - x_0 = \frac{v_0 + v}{2} \cdot \frac{v - v_0}{a}.$$

Use $(v_0 + v)(v - v_0) = v^2 - v_0^2$:

$$x - x_0 = \frac{v^2 - v_0^2}{2a}.$$

Therefore

$$v^2 = v_0^2 + 2a(x - x_0).$$

7.2 Full example: a moving object

An object starts at $x_0 = 0\,\text{m}$, with initial velocity $v_0 = 2\,\text{m/s}$, and acceleration $a = 3\,\text{m/s}^2$. Find its position and velocity after $4\,\text{s}$.

Velocity:

$$v(4) = v_0 + a \cdot 4 = 2 + 3 \cdot 4 = 14\,\text{m/s}.$$

Position:

$$x(4) = x_0 + v_0 \cdot 4 + \frac{1}{2}a \cdot 4^2.$$

Substitute:

$$x(4) = 0 + 2 \cdot 4 + \frac{1}{2} \cdot 3 \cdot 16 = 8 + 24 = 32\,\text{m}.$$

8 Free fall

Near Earth's surface, objects fall with acceleration approximately

$$g \approx 9.81\,\mathrm{m/s^2}.$$

Often in simple problems we use

$$g \approx 10\,\mathrm{m/s^2}.$$

If upward is the positive direction, then gravity points downward, so

$$a = -g.$$

Vertical motion under gravity

If upward is positive:

$$y(t) = y_0 + v_0 t - \frac{1}{2}gt^2,$$

$$v(t) = v_0 - gt.$$

Object dropped from rest

A ball is dropped from height $h = 20\,\mathrm{m}$. It starts from rest, so $v_0 = 0$. Take $g = 10\,\mathrm{m/s^2}$. How long does it take to hit the ground?
Use

$$y(t) = h - \frac{1}{2}gt^2.$$

The ground is $y = 0$, so

$$0 = 20 - \frac{1}{2} \cdot 10 \cdot t^2.$$

Thus

$$0 = 20 - 5t^2,$$

$$5t^2 = 20,$$

$$t^2 = 4,$$

$$t = 2\,\mathrm{s}.$$

Object thrown upward

A ball is thrown upward with initial velocity $v_0 = 20\,\mathrm{m/s}$. Use $g = 10\,\mathrm{m/s^2}$. Find the time to reach maximum height.
At maximum height, velocity is zero:

$$v(t) = v_0 - gt = 0.$$

So

$$20 - 10t = 0,$$

$$t = 2\,\mathrm{s}.$$

The height gained is

$$\Delta y = v_0 t - \frac{1}{2}gt^2 = 20 \cdot 2 - \frac{1}{2} \cdot 10 \cdot 4 = 40 - 20 = 20\,\mathrm{m}.$$

9 Motion in two dimensions

In two dimensions, position is a vector:

$$\boldsymbol{r}(t) = (x(t), y(t)).$$

Velocity is

$$\boldsymbol{v}(t) = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \left(\frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}\right).$$

Acceleration is

$$\boldsymbol{a}(t) = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \left(\frac{\mathrm{d}^2x}{\mathrm{d}t^2}, \frac{\mathrm{d}^2y}{\mathrm{d}t^2}\right).$$

Component-by-component thinking

A two-dimensional motion problem is often just two one-dimensional problems at the same time: one in x, one in y.

9.1 Projectile motion

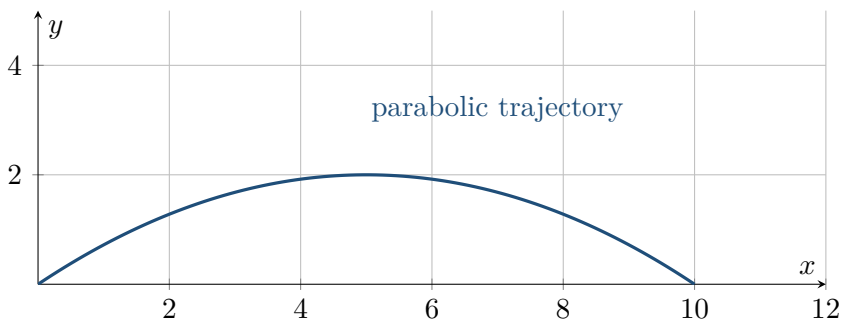
Projectile motion is motion under gravity, ignoring air resistance.

Assume:

- horizontal acceleration is zero: $a_x = 0$,
- vertical acceleration is $-g$: $a_y = -g$,
- initial position is (x_0, y_0),
- initial velocity is (v_{0x}, v_{0y}).

Then

$$x(t) = x_0 + v_{0x}t,$$
$$y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2.$$



9.2 Full example: projectile from ground

A projectile is launched from the ground with

$$v_0 = 20\,\mathrm{m/s}$$

at angle 30° above the horizontal. Use $g = 10\,\mathrm{m/s^2}$. Find the time of flight and range.

Break initial velocity into components:

$$v_{0x} = v_0 \cos 30^\circ = 20 \cdot \frac{\sqrt{3}}{2} = 10\sqrt{3}\,\mathrm{m/s},$$

$$v_{0y} = v_0 \sin 30^\circ = 20 \cdot \frac{1}{2} = 10\,\mathrm{m/s}.$$

Vertical position:

$$y(t) = 10t - 5t^2.$$

The projectile lands when $y(t) = 0$:

$$10t - 5t^2 = 0.$$

Factor:

$$5t(2 - t) = 0.$$

So

$$t = 0 \quad \text{or} \quad t = 2\,\mathrm{s}.$$

The nonzero answer is the flight time:

$$T = 2\,\mathrm{s}.$$

Horizontal position:

$$x(t) = 10\sqrt{3}\,t.$$

Range:

$$R = x(2) = 20\sqrt{3}\,\mathrm{m} \approx 34.6\,\mathrm{m}.$$

10 Forces and Newton's laws

10.1 What is a force?

A force is an interaction that can change the motion of an object. Forces are vectors. They have magnitude and direction.

Examples:

- gravity,
- normal force from a surface,
- friction,
- tension in a rope,
- spring force,
- electric force.

10.2 Newton's first law

Newton's first law

If the net force on an object is zero, the object keeps its velocity constant.

This includes two cases:

- if it is at rest, it remains at rest,
- if it is moving, it keeps moving in a straight line with constant speed.

10.3 Newton's second law

Newton's second law

The net force equals mass times acceleration:

$$\sum \boldsymbol{F} = m\boldsymbol{a}.$$

In one dimension:

$$\sum F = ma.$$

This is the most important equation in elementary mechanics.

What mass means

For the same force, a larger mass has smaller acceleration:

$$a = \frac{F}{m}.$$

Mass measures resistance to acceleration.

10.4 Newton's third law

Newton's third law

If object A exerts a force on object B, then object B exerts an equal and opposite force on object A.

Symbolically:

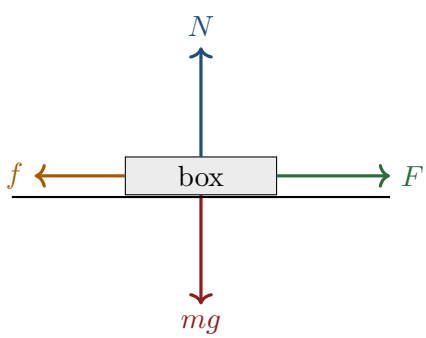
$$\boldsymbol{F}_{A\to B} = -\boldsymbol{F}_{B\to A}.$$

Common beginner mistake

The two forces in Newton's third law act on different objects. They do not cancel when analyzing just one object.

11 Drawing free-body diagrams

A free-body diagram shows all forces acting on one object.



For a box on a table, typical forces are:

- weight mg downward,

- normal force N upward,
- applied force F,
- friction f opposite motion or attempted motion.

11.1 Recipe for Newton problems

Algorithm: solving force problems

1. Choose the object to analyze.
2. Draw all forces acting on that object.
3. Choose coordinate axes.
4. Write Newton's second law in each direction.
5. Solve the equations.
6. Check units and signs.

12 Weight and normal force

Weight is the gravitational force near Earth:

$$W = mg.$$

It points downward.

The normal force is the force from a surface pushing perpendicular to the surface. On a horizontal table, it points upward.

12.1 Object resting on a horizontal table

For an object at rest on a table:

$$a_y = 0.$$

Newton's second law in the vertical direction gives

$$N - mg = 0.$$

So

$$N = mg.$$

Normal force is not always equal to weight

$N = mg$ only in simple cases, such as a body resting on a horizontal surface with no other vertical forces. On an incline, in an elevator, or with extra vertical forces, N can be different.

13 Friction

Friction acts along a surface and opposes relative motion or attempted relative motion.

Two simple models:

- static friction: before sliding starts,
- kinetic friction: during sliding.

Kinetic friction is often modeled as

$$f_k = \mu_k N,$$

where μ_k is the coefficient of kinetic friction.

Static friction satisfies

$$f_s \leq \mu_s N.$$

Static friction adjusts

Static friction is not automatically equal to $\mu_s N$. It can be smaller. The value $\mu_s N$ is the maximum possible static friction.

13.1 Full example: pulling a box

A box of mass $m = 5\,\text{kg}$ is pulled horizontally with force $F = 30\,\text{N}$. The coefficient of kinetic friction is $\mu_k = 0.2$. Use $g = 10\,\text{m/s}^2$. Find the acceleration.

Normal force:

$$N = mg = 5 \cdot 10 = 50\,\text{N}.$$

Kinetic friction:

$$f_k = \mu_k N = 0.2 \cdot 50 = 10\,\text{N}.$$

Net horizontal force:

$$F_{\text{net}} = 30 - 10 = 20\,\text{N}.$$

Newton's second law:

$$F_{\text{net}} = ma.$$

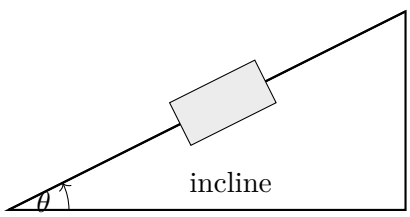
So

$$a = \frac{20}{5} = 4\,\text{m/s}^2.$$

14 Inclined planes

An inclined plane is a classic problem because gravity must be split into components.

Consider a block on a slope of angle θ.



The weight mg points vertically downward. Relative to the slope, it has two components:

$$mg\sin\theta \quad \text{down the slope,}$$

$$mg\cos\theta \quad \text{into the slope.}$$

Therefore, if there is no friction, the acceleration down the incline is

$$a = g\sin\theta.$$

14.1 Full example: frictionless incline

A block slides down a frictionless incline with $\theta = 30°$. Use $g = 10\,\mathrm{m/s^2}$.

Acceleration:

$$a = g \sin 30° = 10 \cdot \frac{1}{2} = 5\,\mathrm{m/s^2}.$$

Important idea

Mass cancels. A heavy block and a light block slide with the same acceleration on a frictionless incline.

15 Work and energy

15.1 Work

Work measures energy transferred by a force during displacement.

If a constant force F acts in the same direction as displacement s, then

$$W = Fs.$$

If force and displacement form an angle θ, then

$$W = Fs\cos\theta.$$

Work by a constant force

$$W = \boldsymbol{F} \cdot \Delta \boldsymbol{r}.$$

The dot product automatically keeps only the part of the force along displacement.

15.2 Kinetic energy

Kinetic energy is energy of motion:

$$K = \frac{1}{2}mv^2.$$

15.3 Work-energy theorem

Work-energy theorem

The net work done on an object equals its change in kinetic energy:

$$W_{\text{net}} = \Delta K = K_f - K_i.$$

15.4 Derivation in one dimension

Start with Newton's second law:

$$F = ma.$$

Use the identity

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x}.$$

Then

$$F = mv\frac{\mathrm{d}v}{\mathrm{d}x}.$$

Multiply by $\mathrm{d}x$:

$$F\,\mathrm{d}x = mv\,\mathrm{d}v.$$

Integrate from initial to final state:

$$\int_{x_i}^{x_f} F\,\mathrm{d}x = \int_{v_i}^{v_f} mv\,\mathrm{d}v.$$

The left side is work. The right side is

$$m\int_{v_i}^{v_f} v\,\mathrm{d}v = m\left[\frac{1}{2}v^2\right]_{v_i}^{v_f} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2.$$

So

$$W = K_f - K_i.$$

Work-energy example

A 2 kg object starts from rest. A constant horizontal force of 10 N acts over 5 m. Ignore friction. Find final speed.
Work:

$$W = Fs = 10 \cdot 5 = 50\,\mathrm{J}.$$

Initial kinetic energy is zero. Therefore

$$K_f = 50\,\mathrm{J}.$$

But

$$K_f = \frac{1}{2}mv^2 = \frac{1}{2}\cdot 2 \cdot v^2 = v^2.$$

So

$$v^2 = 50,$$

$$v = \sqrt{50} \approx 7.07\,\mathrm{m/s}.$$

16 Potential energy and conservation of energy

16.1 Gravitational potential energy

Near Earth:

$$U_g = mgh.$$

Here h is height relative to a chosen zero level.

16.2 Spring potential energy

For an ideal spring:

$$U_s = \frac{1}{2}kx^2,$$

where k is the spring constant and x is displacement from equilibrium.

16.3 Mechanical energy

Mechanical energy is

$$E = K + U.$$

If only conservative forces act, then

$$E = \text{constant}.$$

Conservation of mechanical energy

If no non-conservative forces such as friction remove energy, then

$$K_i + U_i = K_f + U_f.$$

Falling from height using energy

A ball is dropped from height $h = 20\,\text{m}$. Ignore air resistance. Find speed just before impact. Use $g = 10\,\text{m/s}^2$.
Initial state:

$$K_i = 0, \qquad U_i = mgh = m \cdot 10 \cdot 20 = 200m.$$

Final state at ground:

$$U_f = 0, \qquad K_f = \frac{1}{2}mv^2.$$

Conservation of energy:

$$0 + 200m = \frac{1}{2}mv^2 + 0.$$

Cancel m:

$$200 = \frac{1}{2}v^2.$$

Thus

$$v^2 = 400,$$

$$v = 20\,\text{m/s}.$$

Mass often cancels

In many gravity problems without air resistance, mass cancels. That does not mean mass is useless; it means all masses accelerate the same way in the same gravitational field.

17 Momentum and impulse

17.1 Momentum

Momentum is

$$\boldsymbol{p} = m\boldsymbol{v}.$$

In one dimension:

$$p = mv.$$

Momentum is a vector because velocity is a vector.

17.2 Impulse

Impulse is force times time interval:

$$\boldsymbol{J} = \boldsymbol{F}\Delta t$$

for a constant force.

More generally:

$$\boldsymbol{J} = \int \boldsymbol{F}(t)\,\mathrm{d}t.$$

Impulse changes momentum:

$$\boldsymbol{J} = \Delta \boldsymbol{p}.$$

17.3 Conservation of momentum

If no external net force acts on a system, total momentum is conserved:

$$\boldsymbol{p}_{\text{total, initial}} = \boldsymbol{p}_{\text{total, final}}.$$

Perfectly inelastic collision

A cart of mass m moves with speed v. It collides and sticks to another cart of mass $2m$ initially at rest. Find final speed.
Initial momentum:

$$p_i = mv + 2m \cdot 0 = mv.$$

Final combined mass:

$$m + 2m = 3m.$$

Let final speed be V. Final momentum:

$$p_f = 3mV.$$

Conservation of momentum:

$$mv = 3mV.$$

Cancel m:

$$v = 3V.$$

So

$$V = \frac{v}{3}.$$

18 Circular motion

Circular motion means motion along a circle. Even if speed is constant, velocity changes because direction changes.

For uniform circular motion:

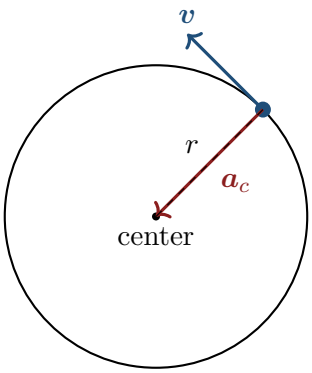
$$a_c = \frac{v^2}{r},$$

where r is radius.

This acceleration points toward the center of the circle. It is called centripetal acceleration.

The required net force is

$$F_c = m\frac{v^2}{r}.$$



Common beginner mistake

In circular motion with constant speed, acceleration is not zero. Velocity changes direction, so acceleration exists.

19 Rotational motion from zero

Rotational motion uses angular quantities.

Linear motion	Rotational motion	Meaning
position x	angle θ	where it is
velocity v	angular velocity ω	how fast it changes
acceleration a	angular acceleration α	how fast velocity changes
mass m	moment of inertia I	resistance to acceleration
force F	torque τ	cause of acceleration

The angular analog of Newton's second law is

$$\tau = I\alpha.$$

For a point mass moving in a circle of radius r:

$$v = r\omega.$$

19.1 Rotational kinetic energy

$$K_{\text{rot}} = \frac{1}{2}I\omega^2.$$

This mirrors linear kinetic energy:

$$K = \frac{1}{2}mv^2.$$

20 Simple harmonic motion

A mass on a spring is the standard example.

Hooke's law:

$$F = -kx.$$

The minus sign means the force pulls back toward equilibrium.

Newton's second law gives

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx.$$

So

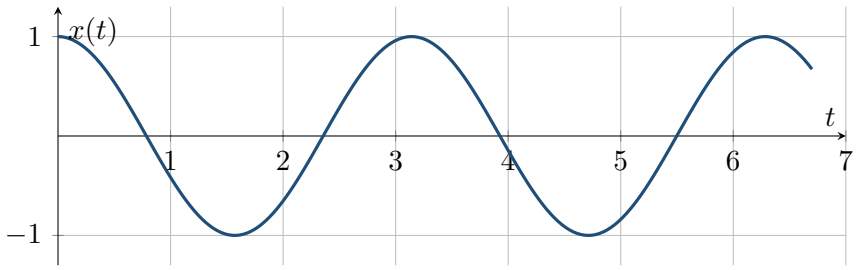
$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{k}{m}x = 0.$$

A solution is

$$x(t) = A\cos(\omega t + \varphi),$$

where

$$\omega = \sqrt{\frac{k}{m}}.$$



Oscillation idea

The force always tries to restore the object to equilibrium. The object overshoots because it has velocity. This repeats, producing oscillation.

21 Numerical simulation for programmers

In programming, we often do not solve equations exactly. We simulate them step by step.

The state of a particle in one dimension can be stored as

$$(x, v).$$

At each time step, we update position and velocity.

21.1 Euler method

Suppose acceleration a is known at the current state. For a small time step Δt:

$$v_{n+1} = v_n + a_n\Delta t,$$

$$x_{n+1} = x_n + v_n\Delta t.$$

This is simple but not very accurate.

Pseudocode: explicit Euler

```
x = initial_position
v = initial_velocity
for each timestep:
    a = force(x, v, t) / m
    x = x + v * dt
    v = v + a * dt
```

21.2 Semi-implicit Euler

Many game engines use a small improvement:

$$v_{n+1} = v_n + a_n \Delta t,$$

$$x_{n+1} = x_n + v_{n+1} \Delta t.$$

The velocity is updated first, then the new velocity is used to update position.

Pseudocode: semi-implicit Euler

```
x = initial_position
v = initial_velocity
for each timestep:
    a = force(x, v, t) / m
    v = v + a * dt
    x = x + v * dt
```

21.3 Why time step size matters

Large Δt causes inaccurate and unstable simulations. Small Δt is more accurate but more expensive computationally.

This is a trade-off between performance and physical correctness.

Simulation warning

A formula may be mathematically correct, but a numerical method may still produce bad results if the time step is too large.

22 A complete simulation example: falling ball

We simulate a ball dropped from height 20 m, with

$$x_0 = 20, \qquad v_0 = 0, \qquad a = -10.$$

Use $\Delta t = 0.5\,\mathrm{s}$ and semi-implicit Euler.

Step	t	a	v after update	x after update
0	0.0	-10	0	20
1	0.5	-10	-5	17.5
2	1.0	-10	-10	12.5
3	1.5	-10	-15	5.0
4	2.0	-10	-20	-5.0

The simulation says the ball crosses the ground between 1.5 s and 2.0 s. The exact solution gives 2 s, but because we used a coarse time step, the simulated position jumps past zero.

23 Choosing the right method

There are often several ways to solve a mechanics problem.

Situation	Good method
Constant acceleration	Kinematic equations
Forces are given	Newton's second law
Height and speed, no friction	Energy conservation
Collision or explosion	Momentum conservation
Programming simulation	Numerical integration
Circular path	Centripetal acceleration
Spring oscillator	Differential equation or energy

Exam strategy

First identify what is conserved or what is known. Do not blindly choose a formula because it looks familiar.

24 Common mistakes and how to avoid them

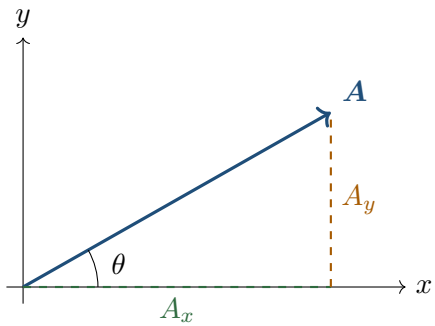
1. **Forgetting units.** Always write units in intermediate steps.
2. **Confusing velocity and acceleration.** Velocity changes position; acceleration changes velocity.
3. **Using speed when direction matters.** Velocity is signed or vector-valued.
4. **Forgetting that forces are vectors.** Add components, not just magnitudes.
5. **Assuming normal force is always mg.** It is not.
6. **Using conservation of energy when friction is present.** Mechanical energy is not conserved with friction.
7. **Mixing degrees and radians.** In calculus and simulations, angles are usually in radians.
8. **Believing simulation equals truth.** Numerical methods approximate the true motion.

25 Vector components without fear

A force or velocity in two dimensions can be split into horizontal and vertical parts. This is called decomposition into components.

If a vector has magnitude A and angle θ measured from the positive x-axis, then

$$A_x = A\cos\theta, \qquad A_y = A\sin\theta.$$



Full example: decomposing a force

A force has magnitude $F = 50\,\mathrm{N}$ and angle $30°$ above the horizontal. Find components.
Horizontal component:

$$F_x = F\cos 30° = 50 \cdot \frac{\sqrt{3}}{2} = 25\sqrt{3}\,\mathrm{N} \approx 43.3\,\mathrm{N}.$$

Vertical component:

$$F_y = F\sin 30° = 50 \cdot \frac{1}{2} = 25\,\mathrm{N}.$$

So

$$\boldsymbol{F} \approx (43.3, 25)\,\mathrm{N}.$$

Sine and cosine confusion

If the angle is measured from the horizontal axis, the horizontal component uses cosine and the vertical component uses sine. If the angle is measured from the vertical axis, the roles swap. Always draw a triangle.

26 Relative motion

Sometimes motion depends on the observer. A person walking inside a moving train has one velocity relative to the train and another velocity relative to the ground.

In one dimension, velocities add:

$$v_{A/C} = v_{A/B} + v_{B/C}.$$

This means:

- velocity of A relative to C,
- equals velocity of A relative to B,
- plus velocity of B relative to C.

Person walking in a train

A train moves at 20 m/s relative to the ground. A person walks forward inside the train at 2 m/s relative to the train. What is the person's velocity relative to the ground?

$$v_{\text{person/ground}} = v_{\text{person/train}} + v_{\text{train/ground}}.$$

So

$$v_{\text{person/ground}} = 2 + 20 = 22\,\text{m/s}.$$

If the person walks backward at 2 m/s, then

$$v_{\text{person/train}} = -2\,\text{m/s},$$

so

$$v_{\text{person/ground}} = -2 + 20 = 18\,\text{m/s}.$$

27 Springs and Hooke's law

A spring is one of the most important simple force models.

Hooke's law

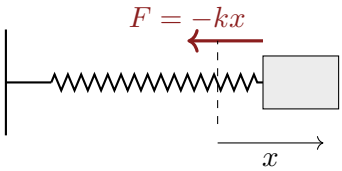
For an ideal spring,

$$F = -kx.$$

Here:

- k is the spring constant,
- x is displacement from equilibrium,
- the minus sign means the force pulls back toward equilibrium.

If $x > 0$, the spring force is negative. If $x < 0$, the spring force is positive. The force always points toward $x = 0$.



Spring force

A spring has $k = 100\,\text{N/m}$. It is stretched by $x = 0.20\,\text{m}$. Find the force.

$$F = -kx = -100 \cdot 0.20 = -20\,\text{N}.$$

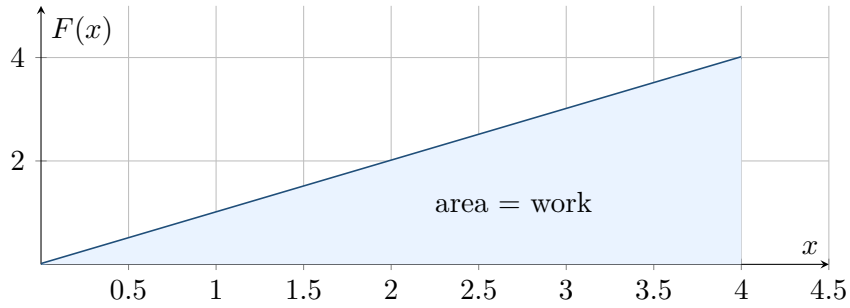
The magnitude is 20 N, and the negative sign means the force points back toward equilibrium.

28 Work by a variable force

For a constant force, work is $W = Fs$. But if force changes with position, we use an integral:

$$W = \int_{x_i}^{x_f} F(x)\,\mathrm{d}x.$$

This is area under the force-position graph.



For a spring, $F = kx$ in magnitude. The work needed to stretch it from 0 to x is

$$W = \int_0^x kx' \, \mathrm{d}x'.$$

We use x' as the integration variable to avoid confusing it with the upper limit.

Compute:

$$W = k \int_0^x x' \, \mathrm{d}x' = k \left[\frac{1}{2} x'^2 \right]_0^x = \frac{1}{2} k x^2.$$

This is why spring potential energy is

$$U_s = \frac{1}{2} k x^2.$$

29 Center of mass

For several point masses on a line, the center of mass is a weighted average of positions:

$$x_{\mathrm{CM}} = \frac{m_1 x_1 + m_2 x_2 + \cdots + m_n x_n}{m_1 + m_2 + \cdots + m_n}.$$

In vector form:

$$\boldsymbol{r}_{\mathrm{CM}} = \frac{m_1 \boldsymbol{r}_1 + m_2 \boldsymbol{r}_2 + \cdots + m_n \boldsymbol{r}_n}{m_1 + m_2 + \cdots + m_n}.$$

Two masses on a line

A mass 2 kg is at $x = 0\,\mathrm{m}$. A mass 6 kg is at $x = 4\,\mathrm{m}$. Find the center of mass.

$$x_{\mathrm{CM}} = \frac{2 \cdot 0 + 6 \cdot 4}{2 + 6} = \frac{24}{8} = 3\,\mathrm{m}.$$

The center of mass is closer to the heavier object.

Why center of mass matters

For many external-force problems, a complicated system moves as if all its mass were concentrated at the center of mass.

30 Torque and static equilibrium

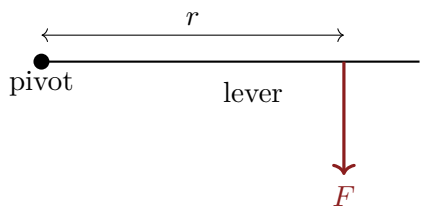
Torque measures the tendency of a force to rotate an object.

For a force perpendicular to a lever arm:

$$\tau = rF.$$

More generally:

$$\tau = rF \sin\theta.$$



For static equilibrium:

$$\sum F = 0, \qquad \sum \tau = 0.$$

Simple lever

A force of 20 N acts downward at distance 3 m from a pivot. What is the torque magnitude?

$$\tau = rF = 3 \cdot 20 = 60\,\mathrm{N\,m}.$$

31 Dimensional analysis as debugging

Programmers debug code. Physicists debug formulas using dimensions.

Suppose we want a formula for the period T of a simple pendulum. The relevant quantities are length L and gravitational acceleration g. Their dimensions are

$$[L] = L, \qquad [g] = LT^{-2}.$$

We need something with dimension time. Try

$$\sqrt{\frac{L}{g}}.$$

Its dimension is

$$\sqrt{\frac{L}{LT^{-2}}} = \sqrt{T^2} = T.$$

So the period must be proportional to

$$T \propto \sqrt{\frac{L}{g}}.$$

The exact small-angle formula is

$$T = 2\pi\sqrt{\frac{L}{g}}.$$

Dimensional analysis cannot give the number 2π, but it gives the correct dependence.

32 Differential equation viewpoint

A differential equation contains an unknown function and its derivatives. Mechanics naturally produces differential equations.

Newton's second law says

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F(x, v, t).$$

This is a rule for acceleration. If we know the initial position and initial velocity, we can compute the motion.

Constant force as differential equation

If $F = F_0$ is constant, then

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F_0.$$

So

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \frac{F_0}{m} = a.$$

Integrating once:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v_0 + at.$$

Integrating again:

$$x(t) = x_0 + v_0t + \frac{1}{2}at^2.$$

Thus the familiar kinematic formulas are solutions of Newton's differential equation.

33 Damping and air resistance: a first model

Real systems often lose energy. A simple model of resistance is a force proportional to velocity:

$$F_d = -bv.$$

Here $b > 0$ is a damping coefficient. The minus sign means the force opposes motion.

For a mass moving horizontally with no other force:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -bv.$$

This says velocity decreases over time.

Without solving the differential equation fully, we can understand the sign:

- if $v > 0$, then $\mathrm{d}v/\mathrm{d}t < 0$, so velocity decreases,
- if $v < 0$, then $\mathrm{d}v/\mathrm{d}t > 0$, so velocity moves back toward zero.

Why damping matters in simulations

Without damping, numerical simulations may look too perfect: objects bounce forever or oscillate forever. Damping is often added to make motion more realistic or more stable.

34 From physics model to code structure

A particle simulator can be organized around the state vector

$$\text{state} = (x, y, v_x, v_y).$$

Forces determine acceleration:

$$a_x = \frac{F_x}{m}, \qquad a_y = \frac{F_y}{m}.$$

Then a numerical method updates the state.

Pseudocode: particle with gravity and drag

```
state = (x, y, vx, vy)
for each timestep:
    Fx = -b * vx
    Fy = -m * g - b * vy

    ax = Fx / m
    ay = Fy / m

    vx = vx + ax * dt
    vy = vy + ay * dt

    x = x + vx * dt
    y = y + vy * dt
```

The physics is in the force model. The numerical method is in the update rule. These are separate design choices.

35 Mini problem set

Problem 1: average velocity

An object moves from $x = 4\,\text{m}$ to $x = 19\,\text{m}$ in $5\,\text{s}$. Find average velocity.

Problem 2: derivative

Position is

$$x(t) = 2t^2 + 3t + 1.$$

Find $v(t)$, $a(t)$, and $v(2)$.

Problem 3: constant acceleration

A car starts from rest and accelerates at $3\,\text{m/s}^2$ for $6\,\text{s}$. Find final speed and distance traveled.

Problem 4: free fall

A stone is dropped from height $45\,\text{m}$. Use $g = 10\,\text{m/s}^2$. Find the time to hit the ground.

Problem 5: Newton's second law

A 4 kg object experiences net force 12 N. Find acceleration.

Problem 6: energy

A 1 kg ball falls from height 5 m. Use $g = 10\,\mathrm{m/s^2}$. Find speed before impact.

Problem 7: collision

A 2 kg cart moving at 6 m/s sticks to a 4 kg cart at rest. Find final speed.

Problem 8: circular motion

A 1 kg object moves in a circle of radius 2 m with speed 4 m/s. Find centripetal acceleration and centripetal force.

36 Solutions to selected problems

Solution 1

$$v_{\text{avg}} = \frac{19-4}{5} = \frac{15}{5} = 3\,\mathrm{m/s}.$$

Solution 2

$$v(t) = x'(t) = 4t + 3.$$
$$a(t) = v'(t) = 4.$$
$$v(2) = 4 \cdot 2 + 3 = 11\,\mathrm{m/s}.$$

Solution 3

$$v = v_0 + at = 0 + 3 \cdot 6 = 18\,\mathrm{m/s}.$$
$$x = x_0 + v_0 t + \frac{1}{2}at^2 = 0 + 0 + \frac{1}{2} \cdot 3 \cdot 36 = 54\,\mathrm{m}.$$

Solution 4

$$0 = 45 - \frac{1}{2} \cdot 10 \cdot t^2.$$
$$0 = 45 - 5t^2.$$
$$5t^2 = 45, \qquad t^2 = 9, \qquad t = 3\,\mathrm{s}.$$

Solution 5

$$a = \frac{F}{m} = \frac{12}{4} = 3\,\mathrm{m/s^2}.$$

Solution 6

$$mgh = \frac{1}{2}mv^2.$$

Cancel m:

$$gh = \frac{1}{2}v^2.$$

$$10 \cdot 5 = \frac{1}{2}v^2.$$

$$50 = \frac{1}{2}v^2, \qquad v^2 = 100, \qquad v = 10\,\mathrm{m/s}.$$

Solution 7

Initial momentum:

$$p_i = 2 \cdot 6 + 4 \cdot 0 = 12.$$

Final mass:

$$2 + 4 = 6\,\mathrm{kg}.$$

Conservation:

$$12 = 6V.$$

Thus

$$V = 2\,\mathrm{m/s}.$$

Solution 8

$$a_c = \frac{v^2}{r} = \frac{4^2}{2} = 8\,\mathrm{m/s^2}.$$

$$F_c = ma_c = 1 \cdot 8 = 8\,\mathrm{N}.$$

37 Formula sheet

Kinematics

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}$$

$$v = \frac{\mathrm{d}x}{\mathrm{d}t}$$

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

Forces

$$\sum \boldsymbol{F} = m\boldsymbol{a}$$

$$W = mg$$

$$f_k = \mu_k N$$

$$f_s \leq \mu_s N$$

Energy

$$W = \boldsymbol{F} \cdot \Delta\boldsymbol{r}$$

$$K = \frac{1}{2}mv^2$$

$$U_g = mgh$$

$$U_s = \frac{1}{2}kx^2$$

$$K_i + U_i = K_f + U_f$$

Momentum

$$\boldsymbol{p} = m\boldsymbol{v}$$

$$\boldsymbol{J} = \Delta\boldsymbol{p}$$

$$\boldsymbol{p}_i = \boldsymbol{p}_f$$

Circular and rotational

$$a_c = \frac{v^2}{r}$$

$$F_c = m\frac{v^2}{r}$$

$$v = r\omega$$

$$\tau = I\alpha$$

$$K_{\text{rot}} = \frac{1}{2}I\omega^2$$

38 Final remarks

Mechanics is not mainly about memorizing formulas. It is about building a model:

1. choose coordinates,
2. identify known quantities,
3. draw forces if needed,
4. write equations,
5. solve step by step,
6. check units and signs,
7. interpret the answer.

For computer science students, mechanics is especially useful because it gives a clear example of how mathematics becomes an algorithm. The real skill is to move between three views:

$$\text{physical idea} \longleftrightarrow \text{mathematical equation} \longleftrightarrow \text{computational update rule.}$$

Once this connection is understood, more advanced topics such as simulation, robotics, control, optimization, graphics, and machine learning for dynamical systems become much easier to approach.